

Performance Testing

Date	30 September 2026
Team ID	SWTID-2026-9986
Project Name	AI-ML & GEN AI – Study Assistant (Smart Study Companion)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how well your system performs under expected load and stress conditions. It helps ensure that the application is fast, stable, and reliable while handling multiple users or requests. Complete the table below with the details of your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter (Load Testing), Locust (Load Testing)
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	Study Assistant – Chat API / Document Retrieval API / User Authentication API
Test Environment	Local / Cloud (AWS EC2) – Staging Environment
Test Date	30 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal Load – User login and chat with AI assistant	50	300	Response time < 3 sec, no errors
2	Peak Load – Multiple users searching & fetching study material	200	600	Response time < 5 sec, stable performance
3	Stress Test – Gradual increase in users	500	900	System handles load without crash
4	Spike Test – Sudden increase in users	1000	300	Graceful degradation, no data loss

5	Database Load – Concurrent read/write operations	300	600	No query timeouts, low error rate
6	Authentication Load – Multiple logins	200	300	Success rate ≥ 99%

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.8 seconds	Pass	Within target
2	Response Time (Max)	< 5 seconds	4.6 seconds	Pass	Within target
3	Throughput (Req/sec)	-	420 req/sec	Pass	Stable
4	Error Rate	< 1%	0.4%	Pass	Acceptable
5	CPU Utilization	< 80%	68%	Pass	Normal
6	Memory Utilization	< 80%	62%	Pass	Normal

Step 4: Observations & Analysis

Key Findings:

- System performed well under normal and peak load conditions.
- Average response time remained below 2 seconds.
- No critical errors were observed during testing.
- Database handled concurrent requests efficiently.

Bottlenecks Identified:

- Slight increase in response time during spike test (4.6 sec) – within acceptable range.

Optimization Steps Taken:

- Enabled caching for frequently accessed study materials.
- Optimized database queries and added connection pooling.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]